Version 8 — Comparative Evaluation Report

Three changes from Version 6, and what Version 8 adds

The caller is the same model in all three versions, sarvam-105b-conversations. Versions 7 and 8 changes when speech starts, when the model is called, and when a call is allowed to end.

The latency benefit

Medians from every saved turn. The mean is in parentheses. Times are seconds.

	Speech to text	Model	Voice	Whole turn
Version 6, all calls	1.03 (1.14)	0.84 (0.86)	3.39 (5.02)	4.58 (6.39)
Version 7, all calls	0.89 (0.90)	0.80 (0.84)	1.26 (1.40)	2.44 (2.56)
Version 8, all calls	1.03 (1.02)	0.90 (0.97)	1.58 (1.66)	2.51 (2.68)
Version 6, four turns or more	0.99 (1.14)	0.80 (0.82)	3.27 (4.75)	4.45 (6.16)
Version 7, four turns or more	0.90 (0.91)	0.80 (0.85)	1.23 (1.38)	2.45 (2.61)
Version 8, four turns or more	0.98 (0.97)	0.80 (0.79)	1.48 (1.52)	2.50 (2.47)

The whole turn falls from 4.58 seconds to 2.44 seconds, then holds at 2.51 seconds. On calls of four turns or more it is 4.45, 2.45, and 2.50 seconds. The Version 6 mean is higher because one voice clip on 12 October at 2:31 PM took 22.6 seconds. The median is the figure to use.

Ishita's pace and warmth

Ishita speaks every line. Version 8 sets her pace and her temperature for each clause. Pace is the speed: 1.0 is her normal rate. A lower pace is slower, and a higher pace is quicker. Temperature is the warmth: a higher value moves more, and a lower value stays even. The speaker stays Ishita.

Line	Pace	Temperature	What the customer hears
Goodbye, or the sales-advisor close	0.9	0.7	A little slower, so the ending is unhurried
A site visit, a day, or a time	1.05	0.65	A little quicker, so the invitation moves forward
Greeting or thank-you	0.95	0.75	The warmest delivery
BHK, size, or price	0.95	0.55	Slightly slower and the steadiest, so a number is easier to catch
Any other line	1.0	0.7	Her normal pace and warmth

A price does not sound like a goodbye, and a visit does not sound like a list of sizes. This is how the line is spoken. The saved voice time remains the moment she starts.

Voice in Version 7 and Version 8 is the moment Ishita starts. Version 6's voice time is the full clip before playback. Speech to text is the same path in all three.

1 Speech starts on the first clause

Version 6	The whole reply is synthesized. Playback starts only after that clip is ready.
Version 7	The first clause plays as soon as it is ready. Later clauses are prepared while that clause is spoken.
Version 8	The same first-clause clock. The median wait before voice is 1.58 seconds, beside 1.26 seconds in Version 7.
Benefit	**From Version 6, speech starts about 2 seconds sooner. Version 8 stays on that clock.**

Change	Benefit
Speech starts on the first clause. The first clause plays as soon as it is ready. Later clauses are prepared while that clause is spoken.	The median wait before the voice falls from 3.39 seconds to 1.26 seconds. The median whole turn falls from 4.58 seconds to 2.44 seconds.

2 Known lines skip the model

Version 6	The model is called on 8 of 15 turns across the three saved calls.
Version 7	The greeting, the configuration question, an open answer, and the visit line are fixed. The 5:45 PM call called the model once in four turns.
Version 8	The model is called on 10 of 57 turns. The 3:56 PM call called it on none of nine turns.
Benefit	**A known reply has no model wait. Version 8 uses that for the price, the open question, and the visit.**

3 The call can close, or book the visit

Version 6	Finishing one topic often produced another question. A no to the project could still be followed by a new offer.
Version 7	A no to the project, a deferred visit, or no details is one goodbye. The clean 5:45 PM call is still in Answer questions.
Version 8	After the price, one open question, then one visit offer. The 3:56 PM call books Saturday at 4:00 PM and closes.
Benefit	**Version 7 can end a call that should end. Version 8 can also book the visit when the customer is ready.**